

A series of standalone product

C ica i c l a al

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1. the defines of data communication packet header structure

```

#define _C_dHd_
    U ig ed h C a d, CheckS , Se i ID, Re l ID;
    TC dHeade , *PC dHeade ;

E la a i :

    C a d: de d
    CheckS : check a d(i cl di g acke heade a d da a)
    Se i ID: e i be
    Re l ID: Re l be :
    Da a acke c e: acke heade + he da a a be a i ed; he achi e
    e e da a acke i a e i h hi acke f c e . Al , he de cha ace (2B e )
+ he check (2B e ) + he e i be (2B e ) + he e e be (2B e ) +
he da a (ce ai le gh) c e a acke T a i a d ecei e acke a e e .

```

the define of command character

Command	Value	Describe
CMD_CONNECT	1000	Connections requests
CMD_EXIT	1001	Di c ec i e e
CMD_ENABLEDEVICE	1002	E e he achi e be a he al k
CMD_DISABLEDEVICE	1003	Make he achi e be a he h -d c di i , ge e all de a e i he k ... LCD
CMD_RESTART	1004	Re a he achi e.
CMD_POWEROFF	1005	Sh -d e ce
CMD_SLEEP	1006	E e he achi e be a he idle a e.
CMD_RESUME	1007	A ake he lee achi e (e a il)

CMD_CAPTUREFINGER	1009	Ca e fi ge i ic e
CMD_TEST_TEMP	1011	Te e fi ge i e i d e
CMD_CAPTUREIMAGE	1012	Ca e he e i e i age
CMD_REFRESHDATA	1013	Ref e h he achi e i e i da a
CMD_REFRESHOPTION	1014	Ref e h he c fig a i a a e e
CMD_TESTVOICE	1017	Pla ice
CMD_GET_VERSION	1100	Ob ai he fi a e edi i
CMD_CHANGE_SPEED	1101	Cha ge a i i eed
CMD_AUTH	1102	C ec i a h i a i
CMD_PREPARE_DATA	1500	P e a e a i he da a
CMD_DATA	1501	T a i a da a acke
CMD_FREE_DATA	1502	Clear machines opened buffer
CMD_DB_RRQ	7	Read i e ki d f da a f he achi e
CMD_USER_WRQ	8	U l ad he e i f a i (f PC e i al).
CMD_USERTEMP_RRQ	9	Read e fi ge i e la e e ki d f da a e i el
CMD_USERTEMP_WRQ	10	U l ad e fi ge i e la e
CMD_OPTIONS_RRQ	11	Read i he achi e e c fig a i a a e e
CMD_OPTIONS_WRQ	12	Se achi e c fig a i a a e e
CMD_ATTLOG_RRQ	13	Read all a e da ce ec d
CMD_CLEAR_DATA	14	Clea da a
CMD_CLEAR_ATTLOG	15	Clea a e da ce ec d

CMD_DELETE_USER	18	Dele e e e
CMD_DELETE_USERTEMP	19	Dele e e fi ge i e la e
CMD_CLEAR_ADMIN	20	Ca cel he a age
CMD_USERGRP_RRQ	21	Read he e g i g.
CMD_USERGRP_WRQ	22	Se e g i g
CMD_USERTZ_RRQ	23	Read he e Ti e Z e e
CMD_USERTZ_WRQ	24	W i e he e Ti e Z e e
CMD_GRPTZ_RRQ	25	Read he g Ti e Z e e
CMD_GRPTZ_WRQ	26	W i e he g Ti e Z e e
CMD_TZ_RRQ	27	Read Ti e Z e e
CMD_TZ_WRQ	28	W i e he Ti e Z e
CMD_ULG_RRQ	29	Read l ck c bi a i
CMD_ULG_WRQ	30	i e l ck c bi a i
CMD_UNLOCK	31	l ck
CMD_CLEAR_ACC	32	Re e Acce C l e he defa l c di i .
CMD_CLEAR_OPLOG	33	Dele e a e da ce achi e all a e da ce ec d.
CMD_OPLOG_RRQ	34	Read a age he ec d
CMD_GET_FREE_SIZES	50	Ob ai achi e c di i ,like e ec di g be a d
CMD_ENABLE_CLOCK	57	E e he achi e be a he al k c di i .
CMD_STARTVERIFY	60	E e he achi e be a he a he ica i c di i
CMD_STARTENROLL	61	S a e ll e e ,e e he achi e be a

		he egi a i e c di i
CMD_CANCELCAPTURE	62	Make he achi e be a he ai i g de a , lea e efe he CMD_STARTENROLL de c i i .
CMD_STATE_RRQ	64	Gai he achi e he c di i
CMD_WRITE_LCD	66	W i e LCD
CMD_CLEAR_LCD	67	Clea he LCD ca i (clea c ee).
CMD_GET_PINWIDTH	69	Ob ai he le g h f e e ial be
CMD_SMS_WRQ	70	U l ad he h e age.
CMD_SMS_RRQ	71	D l ad he h e age
CMD_DELETE_SMS	72	Dele e he h e age
CMD_UDATA_WRQ	73	Se e h e age
CMD_DELETE_UDATA	74	Dele e e h e age
CMD_DOORSTATE_RRQ	75	Ob ai he d c di i
CMD_WRITE_MIFARE	76	W i e he Mifa e ca d
CMD_EMPTY_MIFARE	78	Clea he Mifa e ca d
CMD_GET_TIME	201	Ob ai he achi e i e
CMD_SET_TIME	202	Se achi e i e
CMD_REG_EVENT	500	Regi e he e e
EF_ATTLOG	1	Be eal- i e e if cce f ll
EF_FINGER	(1<<1)	be eal i e e fi ge i (be eal i e e da a e ig)
EF_ENROLLUSER	(1<<2)	Be eal- i e e ll e
EF_ENROLLFINGER	(1<<3)	be eal- i e e ll fi ge i
EF_BUTTON	(1<<4)	be eal- i e e b

EF_UNLOCK	(1<<5)	be eal- i e l ck
EF_VERIFY	(1<<7)	be eal- i e e if fi ge i
EF_FPFTR	(1<<8)	be eal- i e ca e fi ge i i ia
EF_ALARM	(1<<9)	Ala ig al
//F ll i a achi e e de		
CMD_ACK_OK	2000	Re al e f de e f cce f ll
CMD_ACK_ERROR	2001	Re al e f de e f failed
CMD_ACK_DATA	2002	Re da a
CMD_ACK_RETRY	2003	/* Registered event occurred */
CMD_ACK_REPEAT	2004	
CMD_ACK_UNAUTH	2005	C ec i a h i ed
CMD_ACK_UNKNOWN	0xffff	U k de
CMD_ACK_ERROR_CMD	0xfffd	O de fal e
CMD_ACK_ERROR_INIT	0xfffc	/* Not Initializated */
CMD_ACK_ERROR_DATA	0xfffb	
//F ll i da a e ig (a e da ce ec d, fi ge i)		
FCT_ATTLOG	(U8)1	A e da ce ec d
FCT_WORKCODE	(U8)8	WorkCode。
FCT_FINGERTMP	(U8)2	Fi ge i da a
FCT_OPLOG	(U8)4	O e a i ec d
FCT_USER	(U8)5	U e ec d
FCT_SMS	(U8)6	Short message

FCT_UDATA (U8)7 U e h e age

2. The description and detailed explanation to the order characters

CMD_CONNECT. hi de i a lied c ec he achi e, if cceed, he f ll he c e f he acke heade f CMD_ACK_OK e de ,cl ed f ll a acke f c e a i each da a i, acke heade de ailed ea i g de c i i :

Ve ifica i a d (CheckS) alg i h : Acc di g ig h i ege acc la e he e i e acke , ill e 2147483648 (l g 4 b e), gai he l 2 b e al e c i e add ge he agai , de e di g he i i ha he al e i bai ed ge e c le e , a d a f i i he h i ege (ig ed h 2 b e), a el bai he e ifica i .

Se i be (Se i ID): U i el ig a i e f c e c i , e a da ig ed b he achi e he achi e e ec e c e c i

O de e e be (Re l ID): The l a k c e a i i de , i a acc la i al e , e i diffe e ia e he de , a el a f he c e c i , each Re l ID de i diffe e . If he achi e ha e abli hed he c e c i a d , he ca ie cce f ll a f e he c e c i , he achi e e he a h i ed de ha ha c e c ed. Need a i he c e c i a d ce e , ca c le e he c e c i ce , lea e efe he a h i ed de c e c .

◆ CMD_EXIT

Di c ec

◆ CMD_ENABLEDEVICE

E e he achi e be a i he al k c di i , ge e all he da a c ica i hield he achi e a ilia e i e (ke b a d, LCD, e), hi de e e he a ilia e i e be a he al k c di i .

◆ CMD_DISABLEDEVICE

Shield achi e e i he ke b a d, LCD, e , if e f cce f ll , he e a e h i g i he k . LCD.

◆ CMD_CAPTUREFINGER\CMD_CAPTUREFINGER

Ca e i age da a f a achi e e . ZEM200 (A5, F4+ a d) e ie d c d hi f c i . Re e da a i bi a igi ali a k da a , beca e he da a be a i ed i big he ca e bi a , he e i al fi a i a ecial acke ,

he fi 4 b e f hi acke da a a a e acce he al le gh f he da a, he
 he i e f each acke ill be a ed a 4 b e , ce ai l , la he acke i ibl alle
 e al he i e f he acke , al each acke ca e 1Kb e . Whe ead i e
 bigge da a, like all fi ge i e la e, a e da ce ec d , ge e al add a ig i he acke
 f da a a i dica e he da a e ha be ead. If ca e he -c le e ic e, he
 da a a fi 4 b e f he acke i 500 (he ca e a ial i age (all i age)). If ca e
 ic e hich i e i defi ed b i elf, he he da a a a fill i 4 b e id h al e
 f he fi idle b e , beca e he i f he ic e i defi ed, he ef e high d e
 eed be a i ed. he de a i he big i age ha bee e d, he i e
 ecei e he da a (heade) h ld be 640*480, he i e, he da a a 1-4 b e a e bi a
 he DPI,5-8 b e a e bi a id h, 8-12 b e a e bi a ali de

◆ CMD_TEST_TEMP

E a i e e fi ge i e la e e i d e , he da a a a i e fi ge i
 e la e, if he fi ge i e i e ce, e he de cha ac e CMD_ACK_OK,
 The da a a e he e e ial be . The b e c f he e ial be i decided
 b he e e ial be b e hich he achi e ca ..

◆ CMD_REFRESHDATA

Ref e h he da a i he achi e, ai l c le e ch he da a, ef e h he fi ge i
 lib a a d , a d ge e all l ad he a da afe e ec i g hi de .

◆ CMD_REFRESHOPTION

Ref e h he achi e c fig ai , i f he fi a e ec fig e. Like a he achi e
 e ial be , ba d a e, i e a d .

◆ CMD_TESTVOICE

Pla ice. Ma acc di g he cia i add e a d he le gh la ice (l
 ZEM100, beca e ZEM100 e cia i chi), he fi b e f da a a
 f he a i i acke e a a el a i he add e a d he le gh. Al a e he
 i de la ice hich i fi ed cia i .

◆ CMD_CHANGE_SPEED

Cha ge eed he achi e a i i da a. Ta i 0, i dica ed he l d ,
 a i 1, i dica ed he i c ea i g eed.

◆ CMD_AUTH

A h i ed c ec i If he achi e ha e abli hed he ig al c de, he g c ec i g

◆ CMD_PREPARE_DATA

I f he achi e e a ef a i i da a (f e a le PC-> De ice), achi e
 e a ef ecei e da a (De ice-> PC). Fill he le gh f da a hich a be

a i e d i h i a c k e f d a a a f i 4 b e . A f e h e a c h i e e c e i e h i d e , c a a c c d i g e i b e c e a e a b f f e h i c h e e c e i e h e d a a h i c h h a b e e a i e d , a f e e e c i g h i d e c c e f l l , a e h e C M D _ D A T A a i d a a , b e e a i d a a c c e , a i h e d e C M D _ F R E E _ D A T A e l l h e a c h i e h e d a a i c l e e , a d e l e a e h e b f f e a c e e . T h i c e g e e a l a l e e h e f i a e g a d e h e d a a a d i h e b i g d a a c i c a i , c e a i l , i f a c h i e h i d a a l a a l e a f e h e a i i f i i h , e a i l i i i h e d e

◆ C M D _ D B _ R R Q

e a d e a e d d a a c l e e i h e a c h i e i e i , h e d a a i a i e d b h e f i a e d e f i i i c e , f e d e a i l f h e c e l e a e e f e h e f i a e c e f d a a d e f i i i , h e d e a i d a a f l l h e l e h e a d a a i b e a i e d , l e a e e f e h e d e d e c i i f c a i g f i g e i b i a . T h e e e a i d a a i d e c i d e d b h e f i b e f h e a c k e d a a a . I f h e f i b e i a e d a h e i g f h e a e d a c e d a a , h e e a d a l l a e d a c e e c d .
Note: a l a e a d h e b i g d a a d e a l l f l l a b e h e a , l i k e a e d a c e e c d , e , a d f i g e i . O h e e a d h e b i g d a a d e i l l b e l g e e l a e i d e a i l .

◆ C M D _ U S E R _ W R Q

U l a d h e e i f a i ; h e l a d i g d e l h e a i e i e d d a a b a a l l a d h e e i f a i , a d h e f i g e i e l a e , A c c e C l i l e g e a d , f i l l h e d a a a f a c k e d e e d i g h e c e f d a a A f e h e d a a a i i g f i i h , a c c d i g h e e f e d e l h e a c h i e e h e d a a

◆ C M D _ U S E R T E M P _ R R Q

h i d e a e a d e e f i g e i d a a , a i g h e e e i a l b e (2 B e) a d h e f i g e i i d e (0 - 9) h e d a a a f h e a i i a c k e ; T h e d a a a a i g a b e d a a e , h e e a d i e k i d f e c i f i c d a a .

◆ C M D _ U S E R T E M P _ W R Q

U l a d e e f i g e i e l a e , h e a i i a c k e f d a a a c e l e a e e f e h e f i a e c e f d a a .

Note: h e e c d i i c c e f l l l a d h e f i g e i e l a e h i c h i , h e e e i , h e e , h e h e f i g e i i l l b e l a d e d , b e e .

◆ C M D _ O P T I O N S _ R R Q

R e a d i e c f i g a i a a e e a l e i h e a c h i e . N e e d f i l l a c f i g a i a e h e d a a a f h e d a a a c k e , c a i e c c e f l l , e h i c f i g a i a l e .

◆ C M D _ O P T I O N S _ W R Q

Se c fig ai a a e e he achi e. The da a acke da a a eed fill a
c fig ai a e.

◆ CMD_ATTLOG_RRQ

Read all a e da ce ec d (lea e efe he ela i e de c i i ead big da a), hi de
f c i i al e; l a l ead he a e da ce ec d.

◆ CMD_CLEAR_DATA

Clea e ki d f da a, if d a ig he da a e, he dele e all da a, he i e
de e di g he a ig ed e dele e da a

◆ CMD_CLEAR_ATTLOG

Dele e a e da ce ec d

◆ CMD_DELETE_USER

Dele e e e. Fill da a a f he a i i acke i h e e ial be (2B e).

◆ CMD_DELETE_USERTEMP

Dele e e e fi ge i e la e. Fill 1-2 d b e he da a a f a i i acke
i h e e ial be (2B e), fill 3 d b e i h fi ge i e ial be (0-9).

◆ CMD_CLEAR_ADMIN

Dele e he a age

◆ CMD_USERGRP_RRQ

Read he e g . A l ZK Acce C l achi e (F4, F4+, A6 a d)
e a i .fill he fi b e f da a a f he a i i acke i h e e ial
be (2B e).

◆ CMD_USERGRP_WRQ

Se e g . Fill he 1-4 b e f da a a f he a i i acke i h e e ial
be (2B e), 5-8 b e i h he g e ial be .

◆ CMD_USERTZ_RRQ

Read he e e l ed Ti e Z e. Fill 2 b e f he a i i acke da a a i h
e e e ial be (2B e). The e ed 1-4 b e da a i he e e e ial be ; 12 b e
e ec i el i dica e 3 Ti e Z e e ial be hich ed b he e behi d he da e,
each e ial be ake 4 b e .

◆ **CMD_USERTZ_WRQ**

Se e e ed Ti e Z e. Fill 1-4 b e fda a a a i i acke i h ed he Ti e Z e a i , a e e l ca e 3 Ti e Z e, a d he ef e fill i he al e 3. Behi d i 12 b e e a a el fill 3 Ti e Z e e ial be , each ake 3 b e .

◆ **CMD_GRPTZ_RRQ**

Read he g Ti e Z e. Plea e efe ead he e Ti e Z e, he f ci i a e. The de e g Ti e Z e a d e Ti e Z ef ci i i ila a , l ge de c ibe i .

◆ **CMD_TZ_RRQ\CMD_TZ_WRQ**

Read Ti e Z e e i g. Fill he fi 4 b e fda a a f he a i i acke i h Ti e Z e e ial be . Acce C l achi e 50 i eg i a ailable. Each Ti e Z e c clei he eek, i he eek he e i g f f he dail Ti e Z e i 24 h f a , f e a le: 09.091616 i e da Ti e Z e e i g, ea i gal , hi da 08:08 a 16:16 f hi da effec i e Ti e Z e. The e i e Ti e Z e ake he eek a a i , hich acc di g e e da Ti e Z e a a ge. The Ti e Z e a a ge e a f S da .

◆ **CMD_ULG_RRQ\CMD_ULG_WRQ**

Read he g l cki g c bi e. The e i he e e ce e al, a el g c bi a i e i g. Acce C l achi e ca 5 g , 10 l cki g c bi a i a e e . The c bi a i e i e a a ed b ':'. The i hi g c bi a i al i e a a ed b ':'. fill da a a f a i i ece i g acke i h c bi a i i f a i .

◆ **CMD_UNLOCK**

U l ck de . I f he Acce C l achi e l ck, Se he fi 4 b e fda a a he a i i acke a d a i f de la l cki g .

◆ **CMD_GET_FREE_SIZES**

N

◆ **CMD_ENABLE_CLOCK**

Se he LCD d (gli e ':) he acke da a a a i 0 gli e i g, l a gli e . Afe hi de ca ie cce f ll , he fi a e ill ef e h LCD.

◆ **CMD_STARTVERIFY**

E e he achi e be a he e ifica i a e. If he a i i acke ha bee filled

ih e eial be (2B e), he a e if hi e, achi e e e fi ge i. If ha a i ed he e eial be, he achi e a - e e al e ifica i c di i.

◆ CMD_STARTENROLL

Ca e he achi e be a he egi ai a e, fill da a a 1-2 b e f he a i i acke ih e eial be (2B e), 3-4 b e i a lied fill fi ge i eial be, afe e ec i g cce f l LCD e e fi ge i, a egi ai. N e: i he e i g bef e he a i g egi ai, ca e e he egi ai il a i he CMD_CANCEL_CAPTURE de, afe egi e, a e he CMD_START_VERIFY de e e he al e ifica i c di i.

◆ CMD_STATE_RRQ

N

◆ CMD_WRITE_LCD

Thi de a i cha ac e de a e LCD, he da a a 1, 2 b e f he acke a i he a k al e hich a de a e, he 3 db e e i g i 0, f ll cl e he filli g cha ac e hich a be a i. Ma ki CMD_CLEAR_LCD he e hi f ci.

◆ CMD_GET_PINWIDTH

Ob ai he le gh f e eial be, ge e al he eial be i 5 ab e.

◆ CMD_SMS_WRQ

U l ad he h e age. De e di g he h e age c e he da a acke ill be filled, lea e he efe e ce da a c e de ci i.

◆ CMD_SMS_RRQ

D l ad he h e age. Fill he fi 2 b e f da a a f a i i acke ih d l ad he h e age eial be. The e a f he da a acke i e ed b he h e age c e.

Note: The h e age f ci l ca he achi e hich i all ed h e age hi de (f e a le A6).

◆ CMD_DELETE_SMS

Dele e e h e age. Fill he da a a f a i i acke ih 2 b e h e age eial be.

◆ CMD_UDATA_WRQ

Se e h e age. De e di g he e h e age c e fill da a a f he

a i i acke .

◆ CMD_DELETE_UDATA

Dele e e e h e age. Acc di g he h e age c e he da a a f
a i i acke ill be filled i h he e h e age da a a be dele ed.

◆ CMD_DOORSTATE_RRQ

◆ CMD_WRITE_MIFARE

I f achi e i e Mifa e ca d. O l he achi e i h Mifa e ca d eade i able
hi f ci . fill he da a a 1-4 b e f he a i i acke i h e e ial
be , a d 12 b e i h e la e i f ai , he fi 3 b e f he e 12 b e , fill he
1-2 d b e i h e la e le gh , a d he 3 d b e i h e la e i de (c e di g e
e fi ge i), f hi 12 b e , he fi 1-3 b e ca bee , he e be e la e
i he e b e . Ma i e 4 fi ge i e la e , he f ll i g 9 b e e a a el i filled
i h he h e e fi ge i le gh a d he i de , fill e e h e e b e e h d i a e
i h be f e fi lli g 3 b e acki g. Ma e fe he f ll i g c e .
4 h B e [1-2 b e (fi ge i l e la e le gh), 3 d b e (fi ge i i de)],
[4-6... ..] The e be 4 b e fi ge i e la e i f ai 12 b e 1-4
fi ge i e la e da a

◆ CMD_GET_TIME

Ge he i e f he achi e. Fill he da a a 4 b e f he e acke i h he i e
e i g, i e al e i c i e c de, i e c di g e h da f ll : ((Yea %100)*12*31+
((h-1)*31)+da -1)* (24*60*60)+(h *60+ i e) *60+ ec d, acc di g hi
e c di g e h d dec de

◆ CMD_REG_EVENT

Regi e he eal-i e e e . If ha egi e ed hi e e , he achi e ee a i he
eal-i e da a he c eci , like he he e e ifica i i cce fl , kee h ld
he ke b a d , he achi e a i he e i f ai all c eci (cce
c eci).

3. Data structures

Da a ga i a i ha defi i i i he da a c e a i a d ecei e da a acke , i
he defi i i he da a c e i i e a e i h he fi a e e , lea e efe a f ll ,

◆ the user data structure

Bef e he 5.04 edi i fi a e , ha e e he f ll i g da a c e , i if b l
b e .

```
typedef struct _UserOld{
```

```
    U16 PIN;
```

```
    U8 Privilege; // i ilege , 0 , C e , 1 , E lle , 2 , Ma age , 3 ,
```

```
    S e ad i i a
```

```
    char Password[5];
```

```
    char Name[8];
```

```
}TUserOld, *PUserOld;
```

Af e he 5.04 edi i fi a e, he e da a c e a f ll , i if b l b e.

```
typedef struct _User_{
```

```
    U16 PIN; // e e ial be he achi e i e i .
```

```
    U8 Privilege; // ch a f ll i g ill a i g, ega d he 0-7bi, if he 3-1
    bi 000, C e ; 001, E lle ; 110, a age , //, 111, e a age . // he la bi
    f he P i ilege i 0 hich ea i effec , 1, i alid.
```

7	6	5	4	3	2	1	0
0	0	0	0	0	0	1	0

```
//
```

```
//o
```

```
char Password[5]; //Pa d
```

```
char Name[8]; // U e Na e
```

```
U8 Card[5]; //Ca d be , be a l a e he c e di g ID
```

```
Ca d N .
```

```
U8 Group; // e e ec i e g
```

```
U16 TimeZones; // e a e i e Z e, bi ig .
```

```
U32 PIN2; // 32 Bi e ec d ide ifica i be
```



```
}TUser, *PUser;
```

Note: Bef e 5.04 edi i , he achi e l 5 c de ,i de he e Bi e c de,afe 5.04 edi i ha defi ed he e e da a c e, hich e 4 b e i dica e he e e ial be .Na el afe a d 9 bi c de achi e. Whe g c ica i ,bef e i he e a lace he e a e ha e e ial be a i b e , hich i U16 PIN. O he lace a ed U16 PIN i all a e.

◆ **the data structure of fingerprint template**

```
typedef struct _Template_{

    U16 Size; // he le g h f fi ge i e la e

    U16 PIN; // c e d i h he e da a c e PIN

    char FingerID; // fi ge i

    char Valid; // fi ge i i alid i alid

    char *Template; // fi ge i e la e
```

```
}TTemplate, *PTemplate;
```

◆ **Attendance record structure**

The f ll i he a e da ce ec d da a c e f he e e i ec d, a el he a e da ce ec d e he c e i f e .

```
typedef struct _AttLog_{

    Int PIN; //U16 PIN, e e ial be

    char verified; // a chi g a

    time_t time_second; // i e, i e e c di g i c i ed i e e c di g.

    char status; // a e da ce c di i
```

```
}TAttLog, *PAttLog;
```

Note: The ead de l ca be a l he a e da ce ec d, ead i all a e da ce ec d. he c e i f a f he a e da ce ec d i di ided i he le g h h de, i c e i g a a e f ll (f e a le e e l ead i cha *B ffe, he i e i i he fi b e): The fi 2 b e a e ed a e e e ial be (U16 PIN), hi d b e fi h ee (Bi) a e he a e a ch, he 4-5 h a e he a a ch, 6 h bi i ed a e he i dica hich ell h l g i e i; If i i a h i e f , he he i e

al e la l bi f he hi d b e a d f ll b e a e he i e e c i al e, he
a e da ce ec d i e i he i e e c i al e add he i e da al e, he da
i e al e i e ce l al g i e al e (he ef el g de a e i eg i i e, he h
i ef a ed i e e c i al e add he i e da al e), he , igh acc di g he
i e e c di g a (be ible efe c i e e c di g e h d), bai he c ec
i e. F ll i a e e i a e da ce da a ec d c e

```
typedef struct _ExtendAttLog{
```

```
    U32 PIN;
```

```
    time_t time_second; // he e i i eg i i e
```

```
    BYTE status;
```

```
    BYTE verified;
```

```
    BYTE reserved[2]; // e a il i ele .
```

```
    U32 workcode;
```

```
}TExtendAttLog, *PExtendAttLog;
```

If he da a age f i he e e i , he defe hi c e ead he a e da ce
ec d.

◆ short meassage data structure

```
typedef struct _SMS_{
```

```
    U8 Tag; // ca eg , blic h e age, - blic h e age.。
```

```
    U16 ID; // da a c e a ked 0 e e he ec d al ead i alid
```

```
    U16 ValidMinutes; // effec i e i e , 0 = e a e 。
```

```
    U16 Reserved;
```

```
    U32 StartTime; // S a i e
```

```
    U8 Content[MAX_SMS_CONTENT_SIZE+1]; // h e age c e ,
```

```
MAX_SMS_CONTENT_SIZE=60
```

```
}TSms, *PSms;
```

```
U e h e age da a c e
```

```
typedef struct _UData_{
```

```
    U16 PIN; //0 e e i alid ec d .
```

```
    U16 SMSID; // h e age e ial be .
```

```
}TUData, *PUData;
```

◆ management data record structure

```
typedef struct _OPLog_{
```

```
    U16 Admin; // a age NUMBERS
```

```
    BYTE OP; // e a i e
```

```
    time_t time_second; // i e, c le e i e. Ma acc di g c i ed he
```

```
defi i i e h d dec de
```

```
    U16 Users[4]; //User[0], he e e ial be hich i e a ed
```

```
        //. U e [ 1 ], he e a i g e l, he cce i 1, he defea i 0.
```

```
//U e [ 2 ], U e [ 3 ] i ele
```

```
}TOPLog, *POPLog;
```

```
O e a i e de c i i a f ll :
```

The al e	De c ibe
0	T achi e
1	T ff achi e
2	Failed a he ica i a
3	A i-di a le a
4	E e e
5	Cha ge O i
6	Back e ll fi ge i
7	Add Pa d
8	T egi e he HID ca d
9	Dele e U e
10	Dele e fi ge i
11	Dele e Pa d
12	Dele e RF Ca d

13	Clea da a
14	C ea e MF Ca d
15	E ll MF Ca d
16	Regi e MF Ca d
17	Dele e MF ca d egi e d
18	Clea MF Ca d c e
19	T a fe he egi ai da ai he ca d
20	Ca da ai he ca d he a da dal e fi ge i achi e
21	Se he i e f he S a dal e fi ge i achi e
22	e e he lea i g- fac i
23	Clea a e da ce (check-i ,) ec d
24	Clea ad i i a i ilege
25	Re i e Acce C l i
26	Re i e U e Acce C l i
27	Re i e Ti e Z e f Acce C l i
28	Re i e U l ck C bi
29	U l ck
30	E ll U e

4. Real-time events

Afe egi e he e e i he achi e, he achi e i able a i he ela i e eal- i e i f ai he c eci , he ecei e he CMD_REG_EVENT de hich he achi e a i , a el ecei e he eal- i e da a, a acc di g he diffe e da a e ake a al i f he ela ed i f ai . I hi acke hi Se i ID i dica e e e , lea e efe a f ll , fill a d a e i da a a f he ecei ed acke i h ela i e i f ai , a d afe ecei e he eal- i e e age, he e i ID ill cha ge. Need e l he cce f l de .

◆ EF_ATTLOG

Da a a f he e c i ea i ga f ll : 1-2 d b e: U e e ial be ; 4 h b e high 4bi : Whe he effec i e, l 4bi : a e da ce a e; 3 d b e: a chi g a ; The f ll i g 6 b e a e: Whe ea , h a d da i e a d ec d . I iddle- al e i ba ed 2000

◆ EF_ENROLLUSER

Afe egi e e e cce f ll , he achi e e hi eal- i e da a. The da a a f he ecei e acke a bai b e f he e e ial be

◆ EF_ENROLLFINGER

◆ EF_BUTTON

Da a a e al e f he ke b a d ha i e ed.

◆ EF_VERIFY

Da a a e he e e ial be

◆ EF_FPFTR

Da a a e c e he e f he a chi g fi ge i

EF_ALARM

The le g h f he da a a be e ed i 4, he fi b e i 55: a i-di a le ala ; The fi b e i 53: elea e b .

Le g h f he da a a be e ed i 8, he fi b e i 54: cl e d ;

Le g h he da a a be e ed i 12, he fi b e a e 0 FFFF: d e ala 7, 8 b e i he ala e. 5, 6 b e i he d e fi ge i e ial be. 9-12 b e i he a chi g a .

5. Communication Way

◆ UDP

The e i a UDP Se e i 4,370 i he achi e, abili a i ecei e he da a ia UDP c l, he da a i he achi e i c a a i el fe , ead i he g ea e i f a i c e i he fi ge i e la e, he fi ge i e la e ge e all i e ha 600 b e

◆ RS232\485

Pa able RS232\485 c ica i

◆ other

i e achi e , he e i a b il-i WebSe e , a h gh he H e e c ica e. Ne , e achi e a he SOAP c ica i i e face, a e he SOAP c l c ica e. The diffe e achi e ibl he diffe e f c i , l lie he efe e ce.